

Google's new Research : Measuring the environmental impact of delivering AI at Google Scale

Posted by u/Technical-Love-8479 • 0 points • 10 comments

Google has dropped in a very important research paper measuring the impact of AI on the environment, suggesting how much carbon emission, water, and energy consumption is done for running a prompt on Gemini. Surprisingly, the numbers have been quite low compared to the previously reported numbers by other studies, suggesting that the evaluation framework is flawed.

Google measured the environmental impact of **a single Gemini prompt** and here's what they found:

0.24 Wh of energy

0.03 grams of CO₂

0.26 mL of water

Paper :

[https://services.google.com/fh/files/misc/measuring_the_environmental_impact_of_delivering_ai_at_google_scale.pdf]

(https://services.google.com/fh/files/misc/measuring_the_environmental_impact_of_delivering_ai_at_google_scale.pdf)

Video : <https://www.youtube.com/watch?v=q07kf-UmjQo>

Comments

u/danlikendy • 1 points • 2025-08-27 20:09 UTC

So basically one Gemini prompt = one sip of water + a breath of CO2. Feels way too optimistic

u/IronManFolgore • 1 points • 2025-08-27 04:06 UTC

Right now, LLMs are the most expensive they will ever be to train and for inference. they're just going to get cheaper over time. Just like when the first computers came out. Let's see where this goes...

u/Tagwise_ • 1 points • 2025-08-25 15:50 UTC

Surprisingly low compared to other estimations

u/telperion101 • 3 points • 2025-08-25 01:28 UTC

I saw another post where an AI search (water usage specifically) was compared to a pound of ground beef. The beef was still several magnitudes greater but also beef is easier to document it's whole process end to end. the AI lifecycle includes the numerous amount of water used to make the silicon chips, or assembling of the data center - which isn't factored here.

u/Takakeisho5 • 2 points • 2025-08-25 00:54 UTC

Forgive my ignorance but if water is being used for cooling it isn't wasted is it? It can be reused constantly? Or is the water being used for something else

u/Ok_Ad_9986 • 1 points • 2025-08-25 13:19 UTC

It is recycled in a sense but some of it evaporates in each cycle, also heard that it can get contaminated with other chemicals. That "some" is considerable at the scale which they use water.

u/jason-airroi • 0 points • 2025-08-24 15:28 UTC

Yes, the key is their methodology. Most studies just measure the GPU burning energy for your prompt. Google's numbers include all the real-world stuff: idle servers by, cooling, CPU overhead-the whole data center footprint.

So even with that full accounting, the numbers are low. Makes you wonder how efficient their scale actually is vs. older estimates. Just impressive!

u/letsTalkDude • 1 points • 2025-09-02 00:38 UTC

I'm really skeptical of how optimistic numbers are.

u/busybody124 • 19 points • 2025-08-24 11:36 UTC

I'm currently reading the new book *Empire of AI*, and while it's mostly focused on OpenAI, there's a chapter that touches on the controversy of Timnit Gebru's *Stochastic Parrots* paper and her firing from Google. One detail I hadn't heard before was that in the aftermath, Jeff Dean became basically obsessed with showing that Google's energy usage was not as severe as the usage claimed in Strubell (which Gebru had cited and is also the first citation of this paper).

Google is obviously still interested in demonstrating that environmental impact is not as bad as people think, but given that this paper is not peer reviewed, it does sort of border on self-serving PR.

u/darkx0909 • -6 points • 2025-08-24 11:27 UTC

Nice nice

u/DeepAnalyze • 3 points • 2025-08-24 10:58 UTC

Thanks for sharing. This is a crucial piece of the puzzle, but it's important to remember it focuses solely on inference. The paper itself acknowledges that the environmental impact of training large models is the major factor, not serving. While the per-prompt numbers are tiny, they add up over billions of queries. And this is all before we even account for the massive, recurring carbon cost of continuous training and re-training of new models.

u/richizy • 18 points • 2025-08-24 10:32 UTC

0.24 Wh per *median* prompt. They specifically chose the median bc the energy cost distribution is significantly right skewed.

We have no data on whether power users end up using significantly more energy per prompt, e.g. 10x more or even 100x more. Just take a look at how much Google is charging for thinking tokens on Gemini 2.5 Pro. It's significantly more expensive than 2.5 Flash, and I surmise part of the cost is to scale with energy cost.

u/br0monium • 7 points • 2025-08-24 19:52 UTC

That's actually really suspicious because the median holds less information about the aggregate data and holds less predictive power than the mean in this case. We want the total power used, which is simply mean x volume. If we want to forecast expected power usage, that is just mean x expected volume.

The median just says, "the 50% lowest usage prompts use less than this number." Half of all prompts use more energy than the median by definition. If the distribution of power usage has any right skewness at all, then *most* of the power is used by prompts that use more power than the median.

The median doesn't tell us anything about how much more energy the top 50% of prompts use than the bottom. The mean relates to this directly both in calculation (skew and outliers move the mean), and in inference (via the central limit theorem).

u/Bus-cape • 25 points • 2025-08-24 10:23 UTC

I think that's mainly inference, we need to look at the training cost especially knowing that it's not models that we train once, they're always trying to have a better LLM trained with every data they can find each time.